

HOME TASK - ADVANCED SQL QUERIES

Aim

To enhance SQL problem-solving abilities by solving practical database questions using joins, grouping, filtering, subqueries, and aggregate functions.

Objectives

- Understand practical SQL problem-solving techniques.
- Apply JOIN, GROUP BY, HAVING, and aggregate functions.
- Retrieve missing records using LEFT JOIN.
- Use filtering and sorting to generate analytical reports.
- Develop optimized SQL queries for real-world scenarios.

Question 1 (Easy): Matching Skills

Find candidates who possess all the required skills: Python, Tableau, and PostgreSQL.

Solution

```
SELECT candidate_id
FROM candidates
WHERE skill IN ('Python', 'Tableau', 'PostgreSQL')
GROUP BY candidate_id
HAVING COUNT(DISTINCT skill)=3
ORDER BY candidate_id;
```

Question 2 (Medium): Pages With No Likes

Find the IDs of Facebook pages that have never received a like.

Solution

```
SELECT p.page_id
FROM pages p
LEFT JOIN page_likes pl
ON p.page_id=pl.page_id
WHERE pl.page_id IS NULL
ORDER BY p.page_id;
```

Question 3 (Hard): Tesla Unfinished Parts

Find the part names and the number of unfinished parts whose finish date is NULL.

Solution

```
SELECT part,
COUNT(*) AS unfinished_parts
FROM parts_assembly
WHERE finish_date IS NULL
GROUP BY part
ORDER BY unfinished_parts DESC, part;
```

Learning Outcomes

- Write SQL queries using joins and aggregate functions.

- Apply GROUP BY and HAVING for grouped analysis.
- Retrieve unmatched records using LEFT JOIN.
- Solve real-world SQL interview problems efficiently.
- Improve database query writing and optimization skills.